ANNASÈTU (■■■■■■■■) • SIH 2026 Finalist (SIH26032)

Member #5 Profile Dossier

Voice AI & Regional Accessibility Engineer

Focus: Multilingual i18n Engine, Voice Assistant ('■■■■■ ■■■■■') & IVR Phone Simulator

Executive Summary & Core Responsibility:
Created the 5-language localization engine, Web Speech recognition voice assistant, spoken audio synthesis, and IVR dialer.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Localization Engine	Custom Client i18n Engine supporting Hindi, Punjabi, Marathi, Telugu, English
Speech Recognition	Web Speech API (SpeechRecognition interface for Hindi / English input)
Speech Synthesis	SpeechSynthesis API (Configured with native Indian accent, pitch & rate tuning)
Conversational Interface	NLP Intent Extraction for token queries, crop MSP lookups, and queue wait times
IVR Simulation	DTMF Keypad simulation with audio tones, toll-free number simulation

2. Key Modules Developed & Production Deliverables

- **5-Language i18n Localization Engine (app.js):** Implemented instant client-side translation across the entire application with localStorage memory.
- **Interactive Voice Assistant '■■■■■ ■■■■■' (voice_assistant.js):** Built floating voice query assistant allowing farmers to speak queries naturally in Hindi or English.
- **Automated Audio Announcements:** Integrated automatic spoken status readouts (e.g. '■■■■■■■! ■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■■ ■■...') upon stage transitions.
- **Simulated Phone IVR System (ivr_demo.html & voice_ivr.py):** Developed basic-phone IVR simulator with keypad dialing for non-smartphone farmers.
- **Kisan AI Mandi Price Advisor (ai_advisor.html & ai_advisor.py):** Created AI-powered market advisory tool providing crop price trends and MSP comparisons.

3. Direct Codebase Ownership & Modified Files

Source Files:
`frontend/static/js/app.js` • `frontend/static/js/voice_assistant.js` • `frontend/templates/ivr_demo.html` • `frontend/templates/ai_advisor.html` • `backend/app/routers/voice_ivr.py` • `backend/app/routers/ai_advisor.py`

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Demonstrate voice query in Hindi: speak '■■■■ ■■■■ ■■ ■■■■?' and show instant spoken voice reply.
- **Presentation Pitch Point:** Switch languages to Punjabi/Telugu/Hindi and demonstrate full platform translation.
- **Presentation Pitch Point:** Explain the IVR phone solution addressing digital divide and non-smartphone farmer inclusion.